**ANSWER SCHEME**
**QP CODE: 25900093**
**MGU-UGP (HONOURS) REGULAR EXAMINATIONS, MARCH 2025**
**SECOND SEMESTER**
**Discipline Specific Core Course (DSC) – MG2DSCGEO100–EXOGENOUS EARTH PROCESSES**
**(2024 ADMISSION ONWARDS)**

## PART-A
**Answer all questions. Each question carries 1 mark**

1) (b) Condensation - (1 mark) [CO1]
2) (a) Poin bar. (1 mark) [CO3]
3) (d) Erosion at the base of a cliff (1 mark) [CO3]
4) (b) Karst. (1 mark) [CO3]
5) (a) Longshore drift (1 mark) [CO3]

## Part B
**Answer any 7 questions. Each question carries 3 marks (60 words)**

6) Definition of each processe with a short note on geological agents- types of processes (3marks) - Erosion-(1mark) – Transportation (1mark) - Deposition (1mark) [CO3]

7) Soil formation process (0.5 marks) – Influence of parent rock in soil formation - Soil Texture- Mineral Composition of soil and rate of soil formation: Different parent rock – Mafic, Felsic - respond differently to weathering (Basalt weathers faster than Granite). Soil pH: Parent material impacts soil pH; limestone forms alkaline soils, while granite leads to acidic soils. Color and Structure: Parent rock can influence soil color and structure- (Basalt to black soil) - (2.5 marks) [CO2] **(CO3 in QP)**

8) Definition of river terrace (0.5 marks) – Types and their process of formation- Tectonic or Erosion or Climate interaction (1marks) –-Alluvial- Bedrock - Paired – Unpaired Terraces (1.5 marks). [CO3]

9) Definition (1.5 marks) – Erosional landform of glacier – Asymmetrical hillock with a smooth convex side and rugged/step like features on other side - Processes of formation – glacial abrasion on stoss side – plucking on lee side- (1.5 marks) [CO3]

10) Definition (1.5 marks)- Bending of waves-wave in shallower water slow down due to friction with the seabed, while parts in deeper water continue at a faster speed– Significance (1.5 marks) - wave energy distribution – Coastal Erosion and Deposition - sediment transport - coastal straightening by Coastal Erosion and Deposition. [CO3]

11) Sand spits and sand bars are coastal depositional landforms - A sandbar can evolve into a spit when the coastline changes direction (0.5 mark)-figure (0.5 marks). Definition-Spits: narrow, elongated -extend from the shore into the water- often in a hooked shape. (1 mark) Sand bars: wider ridges that run parallel to the coastline and can sometimes block inlets or bays, forming lagoons or sand bar formed by river and types (1 mark) [CO3]

12) Glacial erosion processes with example of features formed: Abrasion (1 mark) – Attrition (1 mark) - Plucking (1 mark) [CO3]

13) - Gaseous envelope, provides Oxygen and CO2 and supports biosphere in the earth system ( 0.5marks) - work as a medium to form clouds and controls rainfall in Hydrosphere - Regulates Temperature -Enables Weather and Climate- Rainfall, storms, and long-term climate patterns - Facilitates Water Cycle (2 marks)- It affects weathering and soil formation of lithosphere (0.5 mark). [CO1]

14) Definition of three main types of wind transportation with figure – Suspension (1 mark) – Saltation (1mark) - traction (1mark) [CO3]

## Part C

15. Products of weathering – Regolith (consist of large and small fragments of rock produced by physical weathering and clay and dissolved ions formed by chemical Weathering. Soil commonly consists of sand, silt, clay, and organic Material) – Soil (Definition) – Soil profile (A-B-C-D Horizons with sketch ad description) – Terrarosa and Domes (Brief description). (4 marks).

Factors:

1. Parent rock - Influence of parent rock in soil formation - Based on Parent rock Texture, colour, structure, fertility, acidity, alkalinity of the soil varies. Example ( 1 mark)
2. Climate variation - Difference in Rate of weathering - different types of soil may form- Example- Pedocal, Caliche, Pedalfer, Laterite, Bauxite. (Any two) -Type of weathering and products of weathering changes based on climatic condition – Insolation in arid terrain- frost wedging in humid terrain. (1.marks)
3. Topography/ slope- hillsides have thin soils and valleys are covered by thicker soil, because soil erodes from hills and accumulates in valleys. (1 mark)
4. Organic activity - Organic activity significantly influences both weathering and soil formation. Biophysical weathering by plants and animals (root wedging or burrowing) - microorganisms and fungi contribute to chemical weathering - organic matter from plant and animal remains, directly impacts soil composition, structure, and fertility. (1 mark)
5. Time- The longer a rock is exposed to the elements (wind, water, temperature changes, etc.), the more it will be weathered and broken down, either physically or chemically. Time acts as a catalyst for weathering and soil formation, allowing for the gradual transformation of rock into fertile soil over geological time scales (1 mark) [CO2]

16. Factors influence stream velocity - **Gradient**- High gradient – fast flow, low gradient -slow flow (2 mark) - **Cross sectional area**- wider channel results slower flow whereas narrow and-deep channel is faster. (2 mark). **Sediment load**- high sediment load causes to reduction in velocity ( 1.5 mark). **Stream discharge**- Stream velocity increases when discharge increases. ( 1.5mark). **Channel roughness**- rough channel creates slow stream flow due to resistance than smooth channel (1 mark) [CO3]

17. Wind as a geomorphic agent significant in arid and semi-arid regions where **vegetation is sparse and the ground is dry and** loose. It shapes the landscape - compared to water and ice, wind has a more **limited and localized impact**- eg. dunes, coast. **Wind's geomorphic role is less significant** than water and ice in the overall shaping of the Earth's surface but wind acts as an **effective transportation agent** especially in arid and semi-arid regions-in **suspension,** fine particles like dust and silt are lifted high into the atmosphere and can be carried over long distances (2 marks). Erosional landforms (blowout, Mushroom rock, Zeugan, yardangs, ventifacts, desert pavement -(3 marks) –Depositional landforms – Sand dunes and types- Loess (2 marks) [CO3]

18. Depositional landforms associated with groundwater activity develop -primarily through the precipitation of minerals from mineral-rich groundwater as it moves through the Earth's crust.( Mark 1) One of the most common settings for these features is in limestone regions, where groundwater dissolves calcium carbonate and later deposits it when conditions change. These formations can result in spectacular underground and surface landscapes, illustrating the significant role of groundwater in shaping landforms (2marks) – process of formation and features of depositional Landforms- Spheleothems - Stalactites-stalagmites-column-flowstone- Travertine Terraces - sinters - (5marks). [CO3]

19) **Air pollution** (contributes to acid rain, which enhances chemical weathering of rocks and buildings) - **Deforestation** ( removes protective vegetation, exposing rocks and soil to the element) - **Mining – quarrying** – improper construction (disturb the land and making them more vulnerable to weathering) - **Agricultural practices** ( such as over-irrigation and the use of chemicals, can alter soil chemistry and promote weathering) - **Urbanization** (disturb surface runoff and water infiltration in nearby rocks) - **human-induced climate change** - global warming (intensifies weathering through more extreme temperature fluctuations and **increased rainfall and melting of snow** - Collectively, these activities contribute to a faster breakdown of the Earth's surface materials.(8 marks –any 8 points) . Increased rate of deforestation reduces root wedging and chemical weathering and hinder weathering [CO2].

[ All COs except the CO against 7 question and Course Code are correct.]

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